Pentium 4 NetBurst Microarchitecture MPCA Module 6

> **Definition:** The **Intel NetBurst Microarchitecture** is a 7th-generation x86 processor design engineered to achieve ultra-high clock speeds (up to 3.8 GHz) and maximum throughput using a deep **20-stage (later 31-stage Prescott) Hyper-Pipelined Technology**.

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1. Detailed Technical Explanation

Functional Components of NetBurst Microarchitecture

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## 2. Deep Dive into NetBurst Innovations

### 1. Execution Trace Cache (L1 I-Cache Replacement)

- Traditional architectures store raw x86 instructions in L1 instruction cache, requiring repeated decode cycles.

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### 2. Hyper-Pipelined Technology (20 to 31 Stages)

- Decomposes instruction processing into 20 distinct pipelined clock stages.

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### 3. Rapid Execution Engine (2x Clock ALUs)

- Integer Arithmetic Logic Units (ALUs) run at **double the core processor frequency**.

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### 4. Quad-Pumped System Bus
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- Transfers data 4 times per clock cycle over a 64-bit bus, delivering up to **6.4 GB/s memory bandwidth** at 800 MHz FSB.

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3. Core Concepts & Memory Keywords

- **NetBurst:** 7th-gen x86 design focused on high clock frequencies.

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4. Quick Recall Flow

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